

Bank Database — ACID Properties & Transaction Isolation

SQL PRACTICAL / LAB EXERCISE

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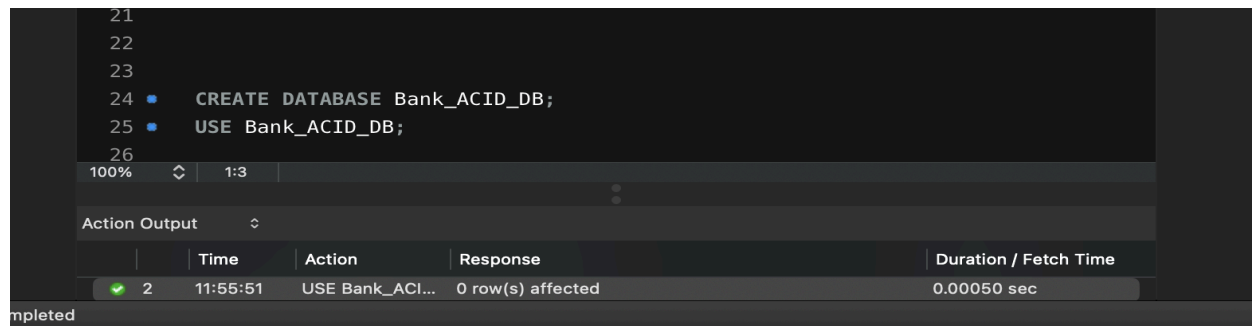
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Section: 7

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Step 1 — Create Database

```
CREATE DATABASE Bank_ACID_DB;  
USE Bank_ACID_DB;
```



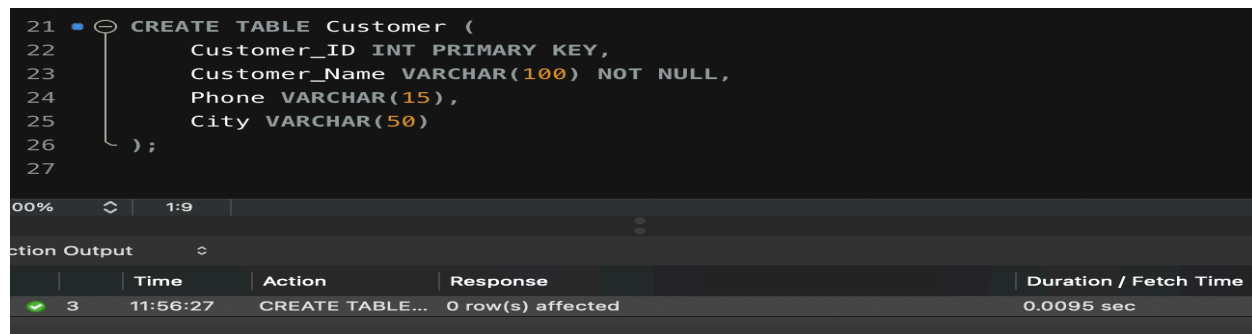
The screenshot shows a SQL IDE interface. The top pane contains the SQL commands: `CREATE DATABASE Bank_ACID_DB;` and `USE Bank_ACID_DB;`. The bottom pane, titled 'Action Output', displays a table with the following data:

	Time	Action	Response	Duration / Fetch Time
2	11:55:51	USE Bank_ACID...	0 row(s) affected	0.00050 sec

A green checkmark is visible next to the first row, indicating successful execution.

Step 2 — Create Customer Table

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    City VARCHAR(50)  
);
```



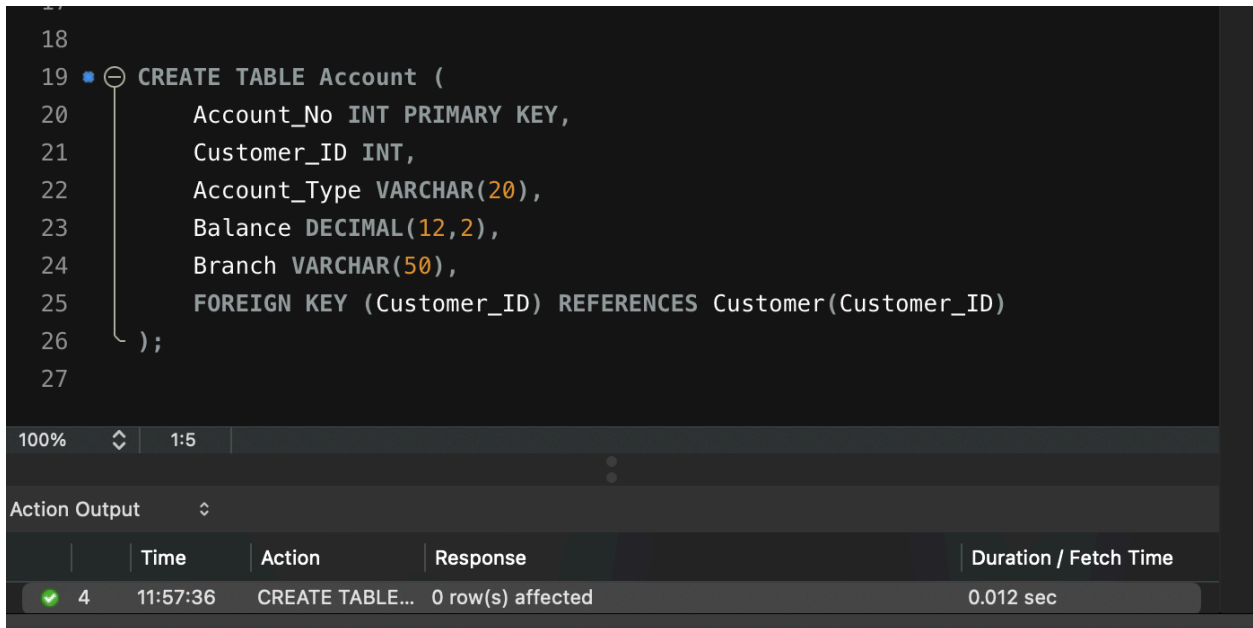
The screenshot shows a SQL IDE interface. The top pane contains the SQL command: `CREATE TABLE Customer (` followed by indented lines for `Customer_ID INT PRIMARY KEY,`, `Customer_Name VARCHAR(100) NOT NULL,`, `Phone VARCHAR(15),`, `City VARCHAR(50)`, and `);`. The bottom pane, titled 'Action Output', displays a table with the following data:

	Time	Action	Response	Duration / Fetch Time
3	11:56:27	CREATE TABLE...	0 row(s) affected	0.0095 sec

A green checkmark is visible next to the first row, indicating successful execution.

Step 3 — Create Account Table (FOREIGN KEY)

```
CREATE TABLE Account (  
    Account_No INT PRIMARY KEY,  
    Customer_ID INT,  
    Account_Type VARCHAR(20),  
    Balance DECIMAL(12,2),  
    Branch VARCHAR(50),  
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
);
```



The screenshot shows a database IDE with a dark theme. The SQL editor displays the same CREATE TABLE statement as in Step 3. Below the editor, the 'Action Output' pane shows a table with columns: Time, Action, Response, and Duration / Fetch Time. The first row indicates that the 'CREATE TABLE...' command was executed successfully at 11:57:36, affecting 0 rows, and taking 0.012 seconds.

```
18  
19 CREATE TABLE Account (  
20     Account_No INT PRIMARY KEY,  
21     Customer_ID INT,  
22     Account_Type VARCHAR(20),  
23     Balance DECIMAL(12,2),  
24     Branch VARCHAR(50),  
25     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
26 );  
27
```

	Time	Action	Response	Duration / Fetch Time
4	11:57:36	CREATE TABLE...	0 row(s) affected	0.012 sec

Step 4 — Insert Customer & Account Data, Then Display

```
INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES  
(101, 'Hansika Reddy', '9876543210', 'Hyderabad'),  
(102, 'Priya Sharma', '9876543211', 'Vijayawada'),  
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),  
(104, 'Sneha Rao', '9876543213', 'Chennai'),  
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');  
  
INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch)  
VALUES  
(10001, 101, 'Savings', 50000, 'Hyderabad'),  
(10002, 102, 'Savings', 75000, 'Vijayawada'),  
(10003, 103, 'Current', 120000, 'Bangalore'),  
(10004, 104, 'Savings', 45000, 'Chennai'),  
(10005, 105, 'Current', 90000, 'Hyderabad');  
  
SELECT * FROM Customer;  
SELECT * FROM Account;
```

35

100% 34:30

Result Grid Filter Rows: Search Edit: Export/Import:

	Account_No	Customer_ID	Account_Type	Balance	Branch
	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Step 5 — Deposit Transaction (COMMIT)

```
START TRANSACTION;
```

```
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
```

```
SELECT * FROM Account WHERE Account_No = 10001;
```

```
COMMIT;
```

```
SELECT * FROM Account WHERE Account_No = 10001;
```

MySQL Workbench

Local instance 3306 - Warning - not supported

Admin... Schemas SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 6* SQL File 7* Context

MANAGEMENT

- Server Stati
- Client Contr
- Users and I
- Status and
- Data Expor
- Data Impor

INSTANCE

Object I... Session

No object selected

1 • START TRANSACTION;

2

3 • UPDATE Account

4 SET Balance = Balance + 5000

5 WHERE Account_No = 10001;

6

7 • SELECT * FROM Account WHERE Account_No = 10001;

8

9 • COMMIT;

10

11 • SELECT * FROM Account WHERE Account_No = 10001;

12

100% 48:7

Result Grid Filter Rows: Search Edit: Export/Import:

	Account_No	Customer_ID	Account_Type	Balance	Branch
	10001	101	Savings	55000.00	Hyderabad
	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Step 6 — Withdraw Transaction (ROLLBACK)

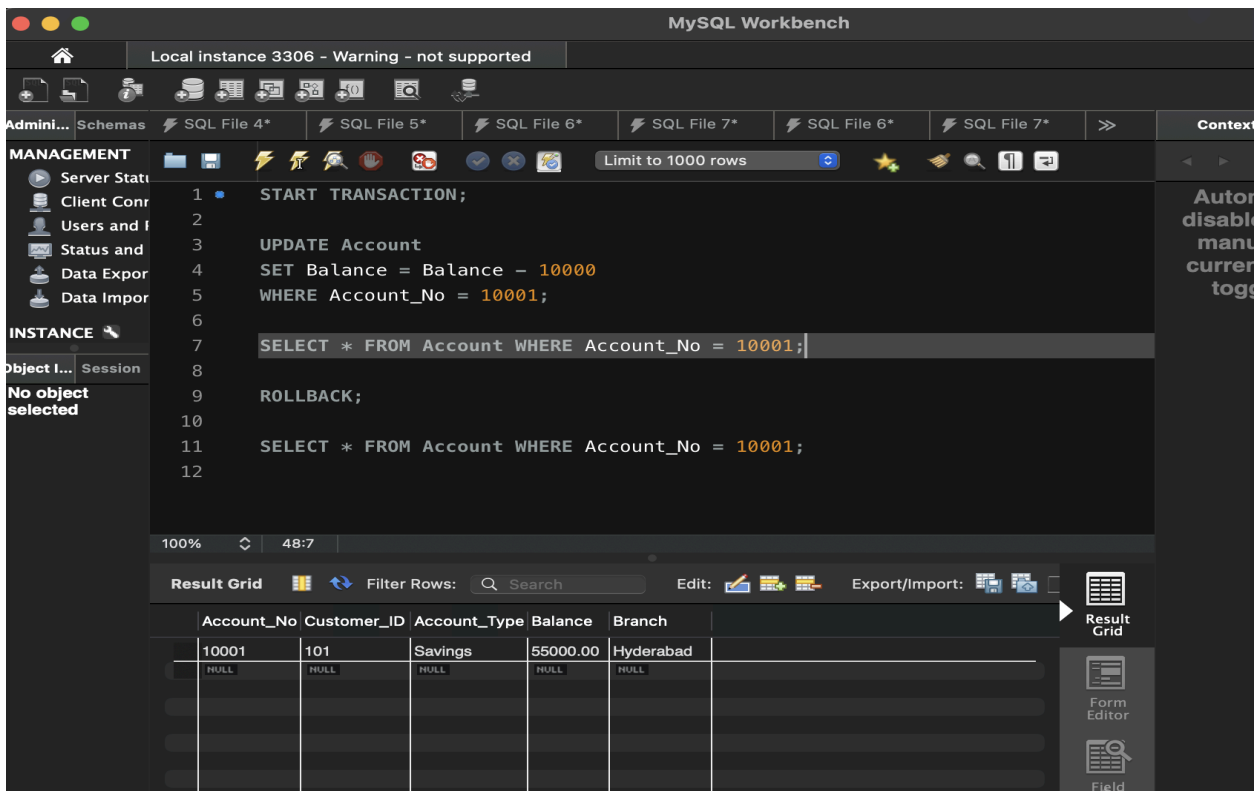
```
START TRANSACTION;

UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;

SELECT * FROM Account WHERE Account_No = 10001;

ROLLBACK;

SELECT * FROM Account WHERE Account_No = 10001;
```



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following transaction:

```
1 START TRANSACTION;
2
3 UPDATE Account
4 SET Balance = Balance - 10000
5 WHERE Account_No = 10001;
6
7 SELECT * FROM Account WHERE Account_No = 10001;
8
9 ROLLBACK;
10
11 SELECT * FROM Account WHERE Account_No = 10001;
12
```

The Result Grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

Step 7 — SAVEPOINT Demonstration

```
START TRANSACTION;

UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001;

SAVEPOINT Deposit1;

UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002;

SAVEPOINT Withdrawal1;
```

```

UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003;

ROLLBACK TO SAVEPOINT Withdrawal1;

COMMIT;

SELECT * FROM Account;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	80000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Step 8 — Bank Transfer (Demonstrates ATOMICITY)

```

START TRANSACTION;

UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001;

UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002;

SELECT * FROM Account WHERE Account_No IN (10001,10002);

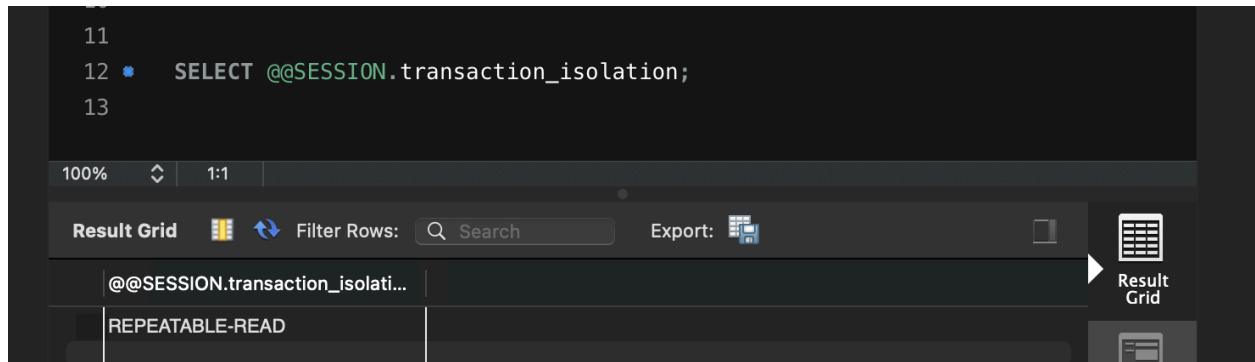
COMMIT;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

Step 9 — Check Current Transaction Isolation Level

```
SELECT @@SESSION.transaction_isolation;
```



Step 10 — Change Isolation Level to READ COMMITTED

```
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;  
  
START TRANSACTION;  
  
SELECT * FROM Account WHERE Account_No = 10001;  
  
COMMIT;
```

